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(54) **LAMINATE FOR HUMIDIFICATION, AND HUMIDIFIER**

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(57) **ABSTRACT**

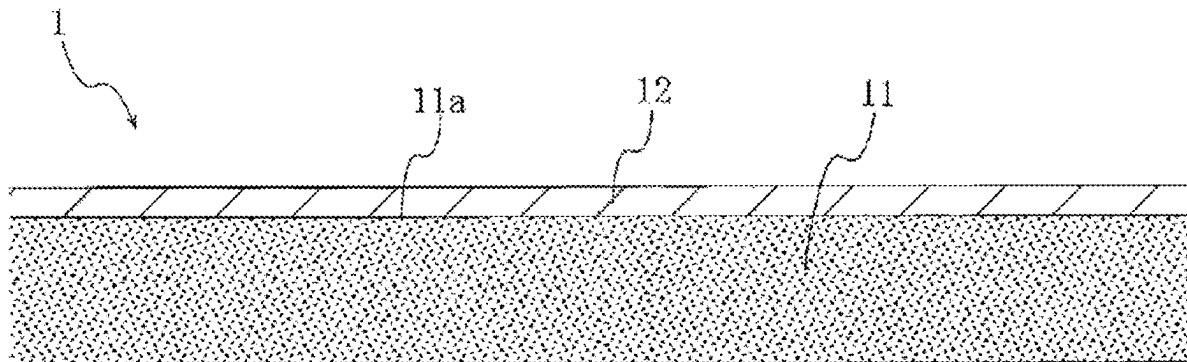
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The present disclosure relates to a laminate for humidification including a porous reinforcing material, and a non-porous membrane that is laminated on at least one surface of the porous reinforcing material and functions as a moisture-permeable membrane.



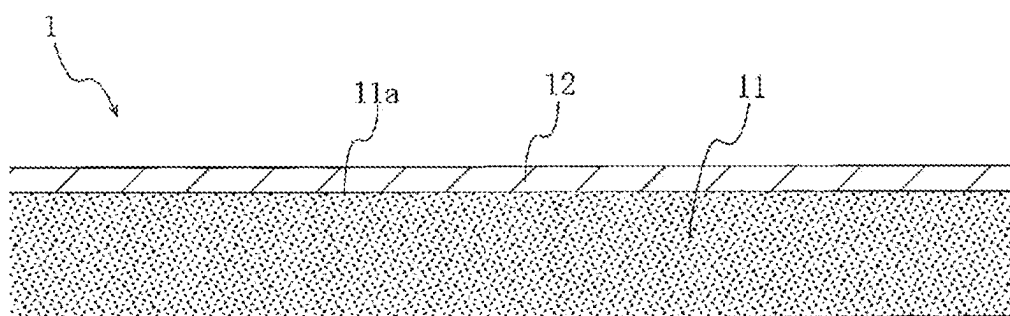


FIG. 1

LAMINATE FOR HUMIDIFICATION, AND HUMIDIFIER

TECHNICAL FIELD

[0001] The present disclosure relates to a laminate for humidification and a humidifier. The present application claims priority to Japanese Patent Application No. 2022-096879 filed in Japan on Jun. 15, 2022, the content of which is incorporated herein by reference.

BACKGROUND ART

[0002] In a moisture-permeable membrane-type humidifier, only water vapor permeates through a moisture-permeable membrane, and air to be humidified and water do not come into direct contact with each other. Thus, scattering of saprophytes and scale components can be suppressed. To improve the moisture permeation performance of such a moisture-permeable membrane-type humidifier, efforts have been made to increase the area of the moisture-permeable membrane. For example, Patent Document 1 discloses a structure in which a three-dimensional porous plate is enclosed in non-porous moisture-permeable membranes.

[0003] The structure including the three-dimensional porous plate as disclosed in Patent Document 1 can improve the moisture permeation performance by increasing the contact surface area between the moisture-permeable membrane and water.

CITATION LIST

Patent Document

[0004] Patent Document 1: JP 8-233315 A

SUMMARY OF INVENTION

Technical Problem

[0005] In the humidifier of Patent Document 1, it is necessary to provide a gap between the non-porous moisture-permeable membranes in which the porous plate is enclosed for achieving air supply properties. This is because when there is no gap, the non-porous membranes may come into contact with each other to cause blocking, and the air supply properties may be impaired. However, the humidifier of Patent Document 1 has a problem that it needs to be increased in size because of the gap provided between the non-porous moisture-permeable membranes. Further, there is also a problem that the cost increases with an increase in the size of the humidifier.

[0006] Thus, an object of the present disclosure is to solve such a problem and to provide a laminate for humidification capable of suppressing blocking between non-porous membranes.

Solution to Problem

[0007] As a result of intensive studies to achieve the above object, the inventors of the present disclosure have found that a laminate for humidification including a porous reinforcing material and a non-porous membrane that is laminated on at least one surface of the porous reinforcing material and functions as a moisture-permeable membrane, the non-porous membrane having water repellency and recesses and protrusions on its surface, can suppress block-

ing between non-porous membranes. The present disclosure relates to what has been completed based on these findings.

[0008] Accordingly, the present disclosure provides a laminate for humidification, the laminate including a non-porous reinforcing material, and a non-porous membrane that is laminated on at least one surface of the porous reinforcing material and functions as a moisture-permeable membrane, wherein a surface of the non-porous membrane has water repellency and recesses and protrusions.

[0009] In the laminate for humidification, the surface of the non-porous membrane has water repellency and recesses and protrusions. Thus, blocking between non-porous membranes can be suppressed.

[0010] The non-porous membrane preferably contains an anti-blocking agent. The inclusion of an anti-blocking agent facilitates suppression of blocking between non-porous membranes.

[0011] Preferably, the anti-blocking agent has an average particle size larger than a thickness of the non-porous membrane, and the recesses and protrusions are formed by the anti-blocking agent. When the average particle size of the anti-blocking agent is larger than the thickness of the non-porous membrane, the surface of the non-porous membrane is readily provided with recesses and protrusions.

[0012] Preferably, a surface of the non-porous membrane has a water contact angle of from 90° to 120°. When the water contact angle of the surface of the non-porous membrane is within the above range, the surface easily exhibits water repellency.

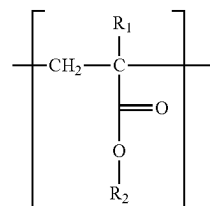
[0013] Preferably, a surface of the non-porous membrane has an arithmetic average roughness (Sa) of from 0.1 μm to 0.5 μm. When the arithmetic average roughness (Sa) of the surface of the non-porous membrane is within the above range, blocking between non-porous membranes can be easily suppressed.

[0014] Preferably, the non-porous membrane is disposed on a liquid water side of the porous reinforcing material and is used to absorb liquid water and release the liquid water into air.

[0015] Preferably, the non-porous membrane contains a thermoplastic resin including a cationic moiety, and the cationic moiety contains a group containing an ammonium ion or a group capable of forming an ammonium ion. When the cationic moiety contains a group containing an ammonium ion or a group capable of forming an ammonium ion, the non-porous membrane can exhibit a bactericidal action, and thus adhesion and deposition of saprophytes or the like can be suppressed.

[0016] Preferably, the non-porous membrane contains a thermoplastic resin including a hydrophilic moiety, and the hydrophilic moiety contains a constituent unit represented by Formula (1) described below. When the non-porous membrane includes the hydrophilic moiety of the aforementioned constituent unit, a water-conducting path can be formed in the non-porous membrane, and moisture permeability can be easily exhibited.

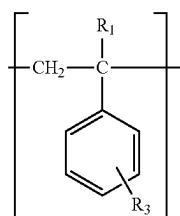
[Chem. 1]



(R₁ and R₂ each independently represent a hydrogen atom or a methyl group.)

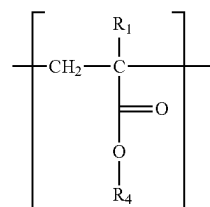
[0017] Preferably, the non-porous membrane contains a thermoplastic resin including a hydrophobic moiety, and the hydrophobic moiety contains a constituent unit represented by Formula (2) and/or Formula (3) described below. When the non-porous membrane includes the hydrophobic moiety of the aforementioned constituent unit, the surface of the non-porous membrane can exhibit water repellency.

[Chem. 2]



[0018] (R₁ represents a hydrogen atom or a methyl group, and R₃ represents a hydrogen atom or an alkyl group having from 1 to 2 carbons.)

[Chem. 3]



[0019] (R₁ represents a hydrogen atom or a methyl group, and R₄ represents a branched alkyl group having 3 or more carbons.)

[0020] The laminate for humidification preferably has a water retention amount of from 5 g/m² to 30 g/m². When the water retention amount is within the above-described range, sufficient moisture permeability can be easily exhibited.

[0021] In the laminate for humidification, preferably, the non-porous membrane is coated to cover at least one surface of the porous reinforcing material.

[0022] The present disclosure also provides a humidifier including a plurality of bag-shaped water-holding containers each including the laminate for humidification and a frame,

the laminate and the frame being bonded together. Since the humidifier includes the laminate for humidification, blocking between non-porous membranes can be suppressed even when the non-porous membranes are disposed close to each other. Thus, the humidifier including the laminate can be easily downsized.

[0023] The present disclosure also provides an air conditioner including the humidifier.

[0024] The present disclosure also provides a ventilator including the humidifier.

[0025] The present disclosure also provides an air purifier including the humidifier.

Advantageous Effects of Invention

[0026] The laminate for humidification of the present disclosure can suppress blocking between non-porous membranes. Accordingly, when the laminate for humidification is used in a humidifier, non-porous membranes can be disposed close to each other, and thus the humidifier can be downsized, resulting in reduced production cost. Therefore, the laminate for humidification of the present disclosure is preferably used in a moisture-permeable membrane-type humidifier.

BRIEF DESCRIPTION OF DRAWINGS

[0027] FIG. 1 is a cross-sectional schematic view illustrating an embodiment of a laminate for humidification of the present disclosure.

DESCRIPTION OF EMBODIMENTS

[Laminate for Humidification]

[0028] A laminate for humidification according to an embodiment of the present disclosure includes at least a porous reinforcing material and a non-porous membrane provided on at least one surface of the porous reinforcing material. The non-porous membrane may be provided on one surface or both surfaces of the porous reinforcing material. The laminate for humidification may have a structure in which the non-porous membrane is sandwiched between two porous reinforcing materials. That is, the porous reinforcing material may be provided on both surfaces of the non-porous membrane. In this case, the two porous reinforcing materials may be the same porous reinforcing material, or may be porous reinforcing materials having different materials, thicknesses, or the like.

[0029] FIG. 1 is a cross-sectional schematic view illustrating an embodiment of the laminate for humidification of the present disclosure. A laminate for humidification 1 includes a porous reinforcing material 11 and a non-porous membrane 12 provided on one surface 11a of the porous reinforcing material 11.

<Non-Porous Membrane>

[0030] The non-porous membrane functions as a moisture-permeable membrane. The surface of the non-porous membrane has water repellency and recesses and protrusions. Thus, when non-porous membranes come into contact with each other, the contact area can be reduced, and adhesion between the non-porous membranes can be prevented. Even when non-porous membranes adhere to each

other, the non-porous membranes can be easily peeled off again, and thus blocking between the non-porous membranes can be prevented.

[0031] The non-porous membrane preferably contains an anti-blocking agent. When the non-porous membrane contains an anti-blocking agent, the surface of the non-porous membrane can be easily provided with recesses and protrusions. Thus, even when non-porous membranes come into contact with each other, blocking between the non-porous membranes can be easily suppressed.

[0032] The average particle size of the anti-blocking agent is preferably from 0.5 μm to 5 μm , more preferably from 1 μm to 4 μm , and particularly preferably from 1.5 μm to 3 μm .

[0033] The thickness of the non-porous membrane is not particularly limited, but is preferably from 0.5 μm to 5 μm , and more preferably from 1 μm to 4 μm . When the thickness is 0.5 μm or more, good membrane formability is achieved, leading to an improvement in a gas barrier property. An increase in the thickness of the non-porous membrane tends to cause more excellent moisture permeability. This is presumed to be because of the high water content of the non-porous membrane. When the thickness is 5 μm or less, a thin membrane is readily formed, resulting in excellent economic efficiency.

[0034] The average particle size of the anti-blocking agent is preferably larger than the thickness of the non-porous membrane, more preferably larger by 0.1 μm to 3 μm , and particularly preferably larger by 0.3 μm to 2 μm . When the average particle size of the anti-blocking agent is larger than the thickness of the non-porous membrane, the surface of the non-porous membrane can be easily provided with recesses and protrusions. From the viewpoint of easy formation of the non-porous membrane, the upper limit of the average particle size is preferably 3 μm or less.

[0035] The recesses and protrusions on the surface of the non-porous membrane are preferably formed by the anti-blocking agent. As long as the surface of the non-porous membrane is provided with recesses and protrusions, the average particle size of the anti-blocking agent may be smaller than the thickness of the non-porous membrane, or the anti-blocking agent may not be contained in the non-porous membrane. When the average particle size of the anti-blocking agent is larger than the thickness of the non-porous membrane, the non-porous membrane can be easily provided with recesses and protrusions. Therefore, when non-porous membranes come into contact with each other, the contact area can be reduced, and thus blocking between the membranes can be suppressed.

[0036] The arithmetic average roughness (Sa) of the surface of the non-porous membrane is preferably from 0.1 μm to 0.5 μm , more preferably from 0.15 μm to 0.45 μm , and particularly preferably from 0.2 μm to 0.4 μm . When the arithmetic average roughness (Sa) of the surface of the non-porous membrane is 0.1 μm or more, the surface of the non-porous membrane can be sufficiently provided with recesses and protrusions. When the arithmetic average roughness (Sa) of the surface of the non-porous membrane is 0.5 μm or less, the non-porous membrane can be easily formed.

[0037] The anti-blocking agent is preferably a fluorine-based, silica-based, or chitosan-based anti-blocking agent. The anti-blocking agent is preferably used in combination with a release agent described below. The use of the anti-blocking agent makes it possible to easily suppress blocking

between the non-porous membrane and the porous reinforcing material. The anti-blocking agent is more preferably a silica-based or chitosan-based anti-blocking agent. One type of the aforementioned anti-blocking agents may be used alone, or two or more types thereof may be used in combination.

[0038] Examples of the fluorine-based anti-blocking agent include triperfluorooctyl phosphate and triperfluorododecyl phosphate. Examples of the silica-based anti-blocking agent include natural silica fine particles and synthetic silica fine particles. Examples of the chitosan-based anti-blocking agent include chitosan nanofibers.

[0039] The content of the anti-blocking agent is preferably from 0.1 parts by mass to 10 parts by mass, more preferably from 0.5 parts by mass to 5 parts by mass, and particularly preferably from 1 part by mass to 3 parts by mass, relative to the entire amount (100 parts by mass) of the copolymer. When the content of the anti-blocking agent is within the above-described range, it is easy to achieve both sufficient anti-blocking performance and moisture permeability.

[0040] The non-porous membrane may contain a release agent. Examples of the release agent include a silicone-based release agent. The release agent exhibits a release action in the non-porous membrane. In the present disclosure, the release action refers to a property of easy re-peeling after adhesion. The use of a material exhibiting the release action causes the non-porous membrane to have water repellency, and the non-porous membranes can be easily peeled off after contacting with each other.

[0041] Examples of the silicone-based release agent include dimethylpolysiloxane. The silicone-based release agent may be modified, and examples of the modified silicone include an amino-modified silicone, an epoxy-modified silicone, and a hydrophilic polyether-modified silicone. One type of the aforementioned release agents may be used alone, or two or more types thereof may be used in combination.

[0042] The content of the release agent is preferably from 0.1 parts by mass to 10 parts by mass, more preferably from 0.5 parts by mass to 5 parts by mass, and particularly preferably from 1 part by mass to 3 parts by mass relative to the entire amount (100 parts by mass) of the copolymer. When the content of the release agent is in the above-described range, it is easy to achieve both sufficient water repellency and moisture permeability.

[0043] The water contact angle of the surface of the non-porous membrane is preferably from 90° to 120°, and more preferably from 92° to 110°. When the water contact angle of the surface of the non-porous membrane is from 90° to 120°, the non-porous membrane can easily exhibit water repellency.

[0044] The non-porous membrane, which is a moisture-permeable membrane, preferably contains a thermoplastic resin to absorb moisture in the non-porous membrane. The thermoplastic resin preferably includes a hydrophilic moiety. The thermoplastic resin preferably includes a hydrophobic moiety to form a water-conducting path in the non-porous membrane and to impart water repellency to the surface of the non-porous membrane. Thus, the thermoplastic resin preferably includes both a hydrophilic moiety and a hydrophobic moiety. It is presumed that when the non-porous membrane includes a structure in which the hydrophilic moiety and the hydrophobic moiety are phase-separated, the hydrophilic moiety functions as a water-

conducting path, and a larger amount of water vapor can be permeated, whereby more excellent moisture permeability is achieved.

[0045] Examples of the thermoplastic resin include acrylic resins, cellulose resins, polyester resins such as polybutylene terephthalate, polyether resins, polyurethane resins, polyvinyl chloride resins, polyethylene, polystyrene resins, polyamide resins, polyacetal resins, polycarbonate resins, polyphenylene sulfide resins, polyether ether ketone, polyimide resins, polytetrafluoroethylene resins, polycaprolactone, and polylactic acid.

[0046] Since the thermoplastic resin preferably includes a hydrophilic moiety and a hydrophobic moiety as described above, the thermoplastic resin is preferably a thermoplastic copolymer containing different monomer components.

[0047] The hydrophilic moiety is preferably composed of a unit of a monomer (a) containing a hydrophilic functional group in a side chain (hereinafter referred to as monomer (a)) among the constituent units of the copolymer. The hydrophobic moiety is preferably composed of a unit of a monomer (b) containing a hydrophobic functional group in a side chain (hereinafter referred to as monomer (b)). The hydrophilic moiety and the hydrophobic moiety are preferably formed in the copolymer. In the non-porous membrane, the copolymer may retain a core-shell structure in which a hydrophobic moiety is formed inside and a hydrophilic moiety is formed outside. In this case, the hydrophilic moiety and the hydrophobic moiety may be formed by a core portion and a shell portion of two or more adjacent copolymers. The copolymer may have a core-shell structure before formation of the non-porous membrane, and may not retain the core-shell structure at the time of forming the non-porous membrane.

[0048] The copolymer preferably contains a structural unit derived from the monomer (a) as a portion constituting the hydrophilic moiety. Examples of the monomer (a) include a glycidyl group-containing monomer, a hydrolyzable silyl group-containing monomer, an acetoacetyl group-containing monomer, a hydroxyl group-containing monomer, a carboxy group-containing monomer, methyl (meth)acrylate, and a monomer having a cationic functional group described below. Among them, the monomer (a) is preferably a carboxy group-containing monomer, methyl (meth)acrylate, and a monomer having a cationic functional group described below. One type of the aforementioned monomers (a) may be used alone, or two or more types thereof may be used in combination. In the present specification, “(meth)acryl” represents at least one of “acryl” and “methacryl”.

[0049] Examples of the glycidyl group-containing monomer include glycidyl (meth)acrylate and glycidyl (meth)allyl ether.

[0050] Examples of the hydrolyzable silyl group-containing monomer include vinyl silyl group-containing monomers such as vinyltrimethoxysilane, vinyltriethoxysilane, vinyltris(β-methoxyethoxy)silane, and vinylmethyldimethoxysilane; and (meth)acryloxy silyl group-containing monomers such as γ-(meth)acryloxypropyltrimethoxysilane, γ-(meth)acryloxypropylmethyldimethoxysilane, γ-(meth)acryloxypropyltriethoxysilane, and γ-(meth)acryloxypropylmethyldiethoxysilane.

[0051] Examples of the acetoacetyl group-containing monomer include diacetoacetic acid allyl ester, acetoacetoxyethyl (meth)acrylate, acetoacetoxyethyl crotonate,

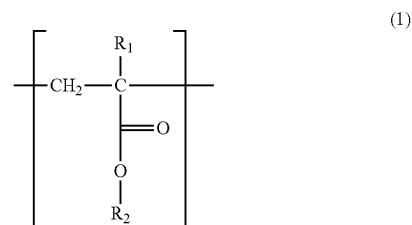
acetoacetoxypropyl (meth)acrylate, acetoacetoxypropyl crotonate, and 2-cyanoacetoacetoxyethyl (meth)acrylate.

[0052] Examples of the hydroxyl group-containing monomer include hydroxyl group-containing (meth)acrylates such as 2-hydroxyethyl (meth)acrylate, hydroxypropyl (meth)acrylate, and 4-hydroxybutyl (meth)acrylate.

[0053] Examples of the carboxy group-containing monomer include crotonic acid, maleic acid, acid anhydride monomers such as maleic anhydride, fumaric acid, citraconic acid, glutaconic acid, itaconic acid, acrylamide N-glycolic acid, cinnamic acid, and (meth)acrylic acid.

[0054] The monomer (a) is particularly preferably methyl (meth)acrylate and/or (meth)acrylic acid, that is, the hydrophilic moiety preferably includes a constituent unit represented by Formula (1) described below.

[Chem. 1]



[0055] (R₁ and R₂ each independently represent a hydro-atom or a methyl group.)

[0056] The content proportion of the monomer (a) is preferably from 20 mol % to 70 mol %, more preferably from 30 mol % to 70 mol %, and still more preferably from 40 mol % to 60 mol % relative to the entire monomer components constituting the copolymer. Adjusting the content proportion of the monomer (a) within this range makes it possible to form a hydrophilic moiety in the non-porous membrane to easily form a water-conducting path, and the non-porous membrane has more excellent moisture permeability.

[0057] The copolymer preferably contains a structural unit derived from the monomer (b) as a portion constituting the hydrophobic moiety. The monomer (b) is not particularly limited, but preferably contains a hydrocarbon group having 2 or more carbons. More preferred examples of the monomer (b) include (meth)acrylic acid esters having a hydrocarbon group having 2 or more carbons. One type of the aforementioned monomers (b) may be used alone, or two or more types thereof may be used in combination.

[0058] Examples of the hydrocarbon group having 2 or more carbons include an aliphatic hydrocarbon group, an alicyclic hydrocarbon group, an aromatic hydrocarbon group, and a group in which two or more of these groups are bound.

[0059] Examples of the aliphatic hydrocarbon group include an alkyl group, an alkenyl group, and an alkynyl group. Examples of the alkyl group include linear or branched alkyl groups such as an ethyl group, a propyl group, an isopropyl group, a butyl group, an isobutyl group, a hexyl group, an octyl group, an isooctyl group, a decyl group, a dodecyl group, and a stearyl group. Examples of the alkenyl group include linear or branched alkenyl groups such as a vinyl group, an allyl group, a methallyl group, a 1-propenyl group, an isopropenyl group, a 1-butenyl group,

a 2-butenyl group, a 3-butenyl group, a 1-pentenyl group, a 2-pentenyl group, a 3-pentenyl group, a 4-pentenyl group, and a 5-hexenyl group. Examples of the alkynyl group include linear or branched alkynyl groups such as an ethynyl group and a propynyl group.

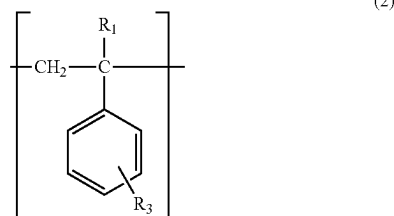
[0060] Examples of the alicyclic hydrocarbon group include cycloalkyl groups having from 3 to 12 carbons, such as a cyclopropyl group, a cyclobutyl group, a cyclopentyl group, a cyclohexyl group, and a cyclododecyl group; cycloalkenyl groups having from 3 to 12 carbons, such as a cyclohexenyl group; and bridged cyclic hydrocarbon groups having from 4 to 15 carbons, such as a bicycloheptanyl group and a bicycloheptenyl group.

[0061] Examples of the aromatic hydrocarbon group include aryl groups having from 6 to 14 carbons (in particular, aryl groups having from 6 to 10 carbons), such as a phenyl group and a naphthyl group.

[0062] The hydrocarbon group having 2 or more carbons is preferably an aliphatic hydrocarbon group and an aromatic hydrocarbon group, and more preferably a linear or branched alkyl group and an aromatic hydrocarbon group.

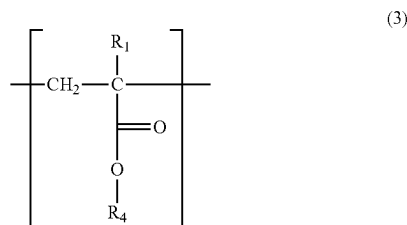
[0063] In particular, the monomer (b) is preferably, for example, a monomer capable of forming a constituent unit described in the following Formula (2) and/or the following Formula (3). That is, the hydrophobic moiety preferably contains a constituent unit represented by the following Formula (2) and/or the following Formula (3).

[Chem. 2]



[0064] (R_1 represents a hydrogen atom or a methyl group, and R_3 represents a hydrogen atom or an alkyl group having from 1 to 2 carbons.)

[Chem. 3]



[0065] (R_1 represents a hydrogen atom or a methyl group, and R_4 represents a branched alkyl group having 3 or more carbons.)

[0066] In Formula (2), R_3 may be bonded at any position of the hydrocarbon constituting the benzene ring, and examples of the monomer capable of forming the constituent

unit represented by Formula (2) include styrene, a-methylstyrene, p-methylstyrene, and p-ethylstyrene.

[0067] In Formula (3), R_4 is preferably a branched alkyl group having from 3 to 10 carbons. Examples of the monomer capable of forming the constituent unit represented by Formula (3) include isopropyl (meth)acrylate, isobutyl (meth)acrylate, tert-butyl (meth)acrylate, and 2-ethylhexyl (meth)acrylate.

[0068] The content proportion of the monomer (b) is preferably from 20 mol % to 70 mol %, more preferably from 30 mol % to 70 mol %, and still more preferably from 40 mol % to 60 mol % relative to the entire monomer components constituting the copolymer. Adjusting the content proportion of the monomer (b) within this range makes it possible to exhibit water repellency and to form a hydrophobic moiety in the non-porous membrane to easily form a water-conducting path, and the non-porous membrane has more excellent moisture permeability.

[0069] The thermoplastic resin preferably includes a cationic moiety. In this case, the thermoplastic resin may include a cationic moiety in the hydrophilic moiety and/or the hydrophobic moiety, or may include a cationic moiety in a portion other than these moieties. The thermoplastic resin preferably includes a cationic moiety in the hydrophilic moiety. The cationic moiety is preferably a cationic functional group or a group capable of forming a cationic functional group. The group capable of forming a cationic functional group refers to a group that does not have a cationic functional group at the stage of the thermoplastic resin but generates a cationic functional group at the stage of forming the non-porous membrane or at the time of using the non-porous membrane. The cationic functional group may be a zwitterionic functional group in which both a cation and an anion are present. Thus, the thermoplastic resin may include an anionic moiety. It is preferable that the thermoplastic resin does not include an anionic moiety.

[0070] Examples of the cationic functional group include a nitrogen-containing group. Examples of the nitrogen-containing group include a group containing an ammonium ion or a group capable of forming an ammonium ion. Here, the group capable of forming an ammonium ion refers to a group that is not ionized at the stage of the thermoplastic resin but forms an ionized body at the stage of forming the non-porous membrane or at the time of using the non-porous membrane. Examples of the group containing an ammonium ion and the group capable of forming an ammonium ion include a primary amino group, a secondary amino group, a tertiary amino group, a quaternary ammonium group, an imino group (—NH— group and =NH group), an amidino group, an imidino group, a hydrazino group, an amide group, an imide group, and a cyclic group containing a nitrogen atom, such as a pyridyl group. Examples of the zwitterionic functional group include a quaternary ammonium-containing group.

[0071] Examples of the quaternary ammonium-containing group include a betaine group and a phosphocholine group. Among them, in the thermoplastic resin, a group capable of generating an ammonium ion after formation of the non-porous membrane is preferred, and a group capable of generating a quaternary ammonium group (quaternary ammonium group, imide group) is more preferred, from the viewpoint of exhibiting antibacterial properties and preventing scattering of saprophytes from the non-porous membrane.

[0072] When the thermoplastic resin is the copolymer, the copolymer preferably contains a structural unit derived from

a monomer having a cationic functional group in a side chain. Examples of the monomer having a cationic functional group include a monomer having the nitrogen-containing group described above as a functional group. Examples of the monomer having the nitrogen-containing group described above as a functional group include an amide group-containing monomer, an amino group-containing monomer, and an imide group-containing monomer. Examples of the amide group-containing monomer include N,N-dialkyl(meth)acrylamides such as N,N-dimethyl(meth)acrylamide, N,N-diethyl(meth)acrylamide, N,N-dipropyl(meth)acrylamide, N,N-diisopropyl(meth)acrylamide, N,N-di(n-butyl)(meth)acrylamide, and N,N-di(t-butyl)(meth)acrylamide; N-alkyl(meth)acrylamides such as N-ethyl(meth)acrylamide, N-isopropyl(meth)acrylamide, N-butyl(meth)acrylamide, and N-n-butyl(meth)acrylamide; N-vinylacetamide; N-hydroxyalkyl(meth)acrylamides such as N-(2-hydroxyethyl)(meth)acrylamide, N-(2-hydroxypropyl)(meth)acrylamide, N-(1-hydroxypropyl)(meth)acrylamide, N-(3-hydroxypropyl)(meth)acrylamide, N-(2-hydroxybutyl)(meth)acrylamide, N-(3-hydroxybutyl)(meth)acrylamide, and N-(4-hydroxybutyl)(meth)acrylamide; N-alkoxyalkyl(meth)acrylamides such as N-methoxymethyl(meth)acrylamide, N-methoxyethyl(meth)acrylamide, and N-butoxymethyl(meth)acrylamide; N,N-dimethylaminopropyl(meth)acrylamide; and N-(meth)acryloylmorpholine. Examples of the amino group-containing monomer include substituted or unsubstituted amino group-containing (meth)acrylates such as aminoethyl (meth)acrylate, N,N-dimethylaminoethyl (meth)acrylate, and t-butylaminoethyl (meth)acrylate. Examples of the imide group-containing monomer include N-(meth)acryloyloxymethylenesuccinimide, N-(meth)acryloyl-6-oxohexamethylenesuccinimide, N-(meth)acryloyl-8-oxohexamethylenesuccinimide, N-cyclohexylmaleimide, N-isopropylmaleimide, N-laurylmaleimide, N-phenylmaleimide, N-methylitaconimide, N-ethylitaconimide, N-butylitaconimide, N-octylitaconimide, N-2-ethylhexylitaconimide, N-cyclohexylitaconimide, and N-laurylitaconimide. One type of the aforementioned monomers having a cationic functional group may be used alone, or two or more types thereof may be used in combination.

[0073] The content proportion of the monomer having a cationic functional group is preferably from 0.5 mol % to 10 mol %, more preferably from 1 mol % to 7 mol %, and still more preferably from 2 mol % to 5 mol % relative to the entire monomer components constituting the copolymer. Adjusting the content proportion of the monomer having a cationic functional group within this range makes it possible to exhibit a repulsive force with the scale component, and adhesion and deposition of the scale component onto the non-porous membrane can be suppressed.

[0074] The copolymer may contain a structural portion derived from a crosslinking agent. The crosslinking agent is not particularly limited as long as it is a compound capable of crosslinking with the side chain of the monomer constituting the copolymer. For example, a bifunctional compound having two reactive functional groups is preferred, and a crosslinking agent having three or more functional groups may be used from the viewpoint of increasing the range of the density of crosslinking in the polymer and adjusting the mechanical properties. Examples of the crosslinking agent include epoxy group-containing compounds having two or more functional groups, and isocyanate group-containing compounds having two or more functional groups. In the

copolymer described above, the crosslinking agent particularly preferably contains an epoxy group-containing compound having two or more functional groups, from the viewpoint that a quaternary ammonium ion can be generated after crosslinking reaction with a functional group formed in a side chain.

[0075] Examples of the epoxy group-containing compound include bisphenol A and epichlorohydrin-type epoxy compounds, polyethylene glycol diglycidyl ether, glycerin diglycidyl ether, glycerin triglycidyl ether, 1,6-hexanediol diglycidyl ether, trimethylolpropane triglycidyl ether, diglycidylaniline, N,N,N',N'-tetraglycidyl-m-xylenediamine (for example, trade name "TETRAD-X", available from Mitsubishi Gas Chemical Company), and 1,3-bis(N,N-diglycidylaminomethyl)cyclohexane (for example, trade name "TETRAD-C", available from Mitsubishi Gas Chemical Company).

[0076] The amount of the crosslinking agent to be used is preferably from 1 part by mole to 60 parts by mole, more preferably from 2 parts by mole to 30 parts by mole, and still more preferably from 4 parts by mole to 15 parts by mole, relative to the total amount (100 parts by mole) of the side chains capable of reacting in the copolymer. Adjusting the amount of the crosslinking agent to be used within this range makes it possible to easily form a water-conducting path in the non-porous membrane, and the non-porous membrane is easily provided with excellent moisture permeability.

[0077] The weight average molecular weight of the thermoplastic resin is not particularly limited, but is preferably from 20000 to 2000000, more preferably from 30000 to 1500000, still more preferably from 50000 to 1000000, and particularly preferably from 70000 to 500000. The weight average molecular weight refers to a value measured by gel permeation chromatography (GPC) and calculated in terms of polystyrene.

[0078] The non-porous membrane may contain an additional component other than the thermoplastic resin, as long as the effects of the laminate for humidification of the present disclosure are not impaired.

[0079] The non-porous membrane may contain an additional component such as a preservative or a leveling agent. The content of the additional component is preferably from 0.1 parts by mass to 10 parts by mass, more preferably from 0.15 parts by mass to 3 parts by mass, and particularly preferably from 0.2 parts by mass to 2 parts by mass, relative to the total amount (100 parts by mass) of the thermoplastic resin.

[0080] Since the copolymer includes a hydrophilic moiety and a hydrophobic moiety, the non-porous membrane has a structure in which the hydrophilic moiety and the hydrophobic moiety are phase-separated at the surface. The maximum diameter of the hydrophilic moiety at the surface of the non-porous membrane is preferably 50 nm or less, more preferably 20 nm or less. When the maximum diameter of the hydrophilic moiety is 50 nm or less, a substance having a size larger than 50 nm is less likely to pass through the non-porous membrane, and the non-porous membrane can be used as a barrier film that does not allow a substance (for example, virus) having a size of 50 nm or more to pass through. The diameter of the hydrophilic moiety is determined by the following method. The diameter of the hydrophilic moiety can be calculated as a circle-equivalent diameter by converting a portion having a high adsorption force (hydrophilic moiety) and a portion having a low adsorption

force (hydrophobic moiety) into numerical values by an adhesion force using an adhesion force measurement mode of a scanning probe microscope (SPM) and processing the values with image analysis software. The maximum diameter corresponds to the largest diameter among the diameters (circle-equivalent diameters) of the hydrophilic moieties calculated as described above.

[0081] In the present disclosure, the structure in which the maximum diameter of the hydrophilic moiety is 50 nm or less does not substantially permeate a component having a size of 50 nm or more, and thus substantially permeates only water vapor. Thus, the pore of the non-porous membrane in the present disclosure excludes a pore having a size of 50 nm or less formed in the hydrophilic moiety. When the maximum diameter of the hydrophilic moiety is larger than 50 nm, there is a possibility that components other than water vapor are permeated. Thus, a membrane having a plurality of structures in which the maximum diameter of the hydrophilic moiety is larger than 50 nm is defined as a porous membrane.

[0082] The non-porous membrane is preferably a layer formed by coating. Coating enables the non-porous membrane to be easily formed, and scattering of scale components, saprophytes, and the like can be easily suppressed.

<Porous Reinforcing Material>

[0083] The above-described porous reinforcing material is an element serving as a support for the non-porous membrane and preferably has excellent moisture permeability.

[0084] The material for forming the porous reinforcing material may be either a hydrophilic material or a hydrophobic material, but is preferably a hydrophobic material. When the hydrophobic material is used, the composition for forming the non-porous membrane does not penetrate into the porous reinforcing material at the time of applying the composition, and thus a lead base material is not required for preventing the composition from flowing off the surface of the porous reinforcing material opposite to the surface on which the coating film is formed.

[0085] Examples of the material for forming the porous reinforcing material include organic materials such as a polyolefin-based resin, a cellulose-based resin, a polycarbonate resin, a polyamide resin, a polyimide resin, a polyamideimide resin, and a fluorine-based resin; and inorganic materials such as metals, glasses, and ceramics. Among them, an organic material is preferred, and the organic material is particularly preferably a polyolefin-based resin, from the viewpoint that the non-porous membrane can be formed on the porous reinforcing material at a relatively low temperature, and from the viewpoint of excellent moisture permeability and water resistance. The aforementioned material may be in the form of fibers such as metal fibers or inorganic fibers. One type of the aforementioned materials for forming the porous reinforcing material may be used alone, or two or more types thereof may be used.

[0086] Examples of the porous reinforcing material include a resin porous membrane, an inorganic porous membrane, a metal porous membrane, and a fibrous base material.

[0087] The polyolefin-based resin is a polymer containing an olefin as an essential monomer component (including an olefin-based elastomer), that is, a polymer containing at least a constituent unit derived from an olefin in a molecule (in one molecule). The olefin is not particularly limited, and

examples thereof include α -olefins such as ethylene, propylene, 1-butene, and 4-methyl-1-pentene.

[0088] Examples of the polyolefin-based resin include a polymer containing ethylene as an essential monomer component (polyethylene-based resin), a polymer containing propylene as an essential monomer component (polypropylene-based resin), an ionomer, and an amorphous cyclic olefin-based polymer.

[0089] The porosity of the porous reinforcing material is not particularly limited, but is preferably from 30 vol % to 90 vol %, more preferably from 40 vol % to 70 vol %. When the porosity is 30 vol % or more, the moisture permeability is further improved. When the porosity is 90 vol % or less, the performance for supporting the non-porous membrane is further improved.

[0090] The thickness of the porous reinforcing material is not particularly limited, but is preferably 5 μ m or more, and more preferably 10 μ m or more from the viewpoint of being able to sufficiently support the non-porous membrane. The thickness of the porous reinforcing material is preferably 50 μ m or less, and more preferably 30 μ m or less from the viewpoint of excellent moisture permeability and in consideration of economic efficiency.

[0091] The surface of the porous reinforcing material on which the non-porous membrane is provided (for example, the surface 11a illustrated in FIG. 1) is preferably subjected to a hydrophilization treatment from the viewpoint of facilitating formation of the non-porous membrane. In particular, when a hydrophobic material is used as the material for forming the porous reinforcing material, the hydrophilization treatment is preferably performed. Examples of the hydrophilization treatment include corona discharge treatment and plasma treatment. Such a hydrophilization treatment makes it possible to generate a carboxy group, a hydroxyl group, or a carbonyl group on the surface of the porous reinforcing material, and the composition for forming the non-porous membrane easily wets and spreads on the surface of the porous reinforcing material, thereby facilitating formation of the non-porous membrane. This also improves the adhesion between the porous reinforcing material and the non-porous membrane. When the porous reinforcing material formed of a hydrophobic base material is stored in the form of a wound body, one surface and the other surface of the porous reinforcing material are in contact with each other in the wound body. However, since one surface (hydrophilic surface) and the other surface (hydrophobic surface) are in contact with each other, blocking can be suppressed.

[0092] The surface tension of the surface of the porous reinforcing material on the side on which the non-porous membrane is formed is preferably from 35 dyn/cm to 55 dyn/cm, more preferably from 37 dyn/cm to 50 dyn/cm. When the surface tension is 35 dyn/cm or more, the composition for forming the non-porous membrane can be easily applied, and the non-porous membrane can be easily formed. When the surface tension is 55 dyn/cm or less, the composition for forming the non-porous membrane does not excessively wet or spread, and the non-porous membrane can be easily formed on the surface of the porous reinforcing material. When the surface of the porous reinforcing material is hydrophilized, the surface on which the non-porous membrane is to be formed is the hydrophilized surface.

[0093] The surface tension of the inside of the porous reinforcing material (i.e., the inside being the region where

the non-porous membrane is not formed) is preferably less than 35 dyn/cm, more preferably 33 dyn/cm or less. When the surface tension is less than 35 dyn/cm, the composition for forming the non-porous membrane is prevented from penetrating to the inside of the porous reinforcing material, and the non-porous membrane can be easily formed on the surface of the porous reinforcing material. When the surface of the porous reinforcing material is subjected to a hydrophilization treatment, the inside of the porous reinforcing material corresponds to the region not subjected to the hydrophilization treatment. The surface tension of the inside can be determined by measuring the surface tension of a cross-section obtained by cutting the porous reinforcing material.

[0094] In the laminate for humidification, it is preferable that blocking is suppressed between non-porous membranes, between porous reinforcing materials, or between the non-porous membrane and the porous reinforcing material when the sheet in a roll form is heated at 70° C. for 15 hours and then pulled out by hand. Particularly in the case of a laminate for humidification in which a non-porous membrane is formed on one side of a porous reinforcing material, it is preferable that blocking is suppressed between the non-porous membrane and the porous reinforcing material. Since blocking is suppressed in each of the above-described modes in the laminate for humidification, the laminate can be easily produced and stored in the form of a wound body.

[0095] The laminate for humidification including the non-porous membrane on at least one surface is formed into two strips each having a size of 25 mm×100 mm. Water is added dropwise to the non-porous membrane of one strip of the laminate for humidification, and then the non-porous membrane is added to the non-porous membrane of the other strip of the laminate for humidification, whereby the entire bonded surface is impregnated with water. When the bonded strips of the laminate for humidification are heated at 55° C. for 1 hour and the peel strength is measured with Tensilon, the peel strength is preferably 3 N/25 mm or less, more preferably 2 N/25 mm or less, and still more preferably 0.5 N/25 mm or less. When the peel strength is 3 N/25 mm or less, even if non-porous membranes come into contact with each other, the non-porous membranes are easily peeled off again, and blocking is easily suppressed.

[0096] The water retention amount of the laminate for humidification is preferably 8 g/m² or more, more preferably 15 g/m² or more, and particularly preferably 15 g/m² or more. In the present disclosure, the water retention amount is determined as follows: the laminate for humidification is cut into a circular test piece having a diameter of 7 cm, the non-porous membrane side of the test piece is impregnated with pure water for 15 minutes, and then the test piece is taken out and weighed. The water retention amount refers to the amount of increase of the laminate for humidification per 1 m² of test piece. When the water retention amount of the laminate for humidification is 8 g/m² or more, the laminate easily exhibits sufficient moisture permeability.

[Method for Producing Laminate for Humidification]

[0097] The laminate for humidification of the present disclosure can be produced by forming the non-porous membrane on at least one surface of the porous reinforcing material by a known or commonly used method. For example, the non-porous membrane may be directly formed on one surface of the porous reinforcing material. Alternatively,

the non-porous membrane may be temporarily formed on another support, and then the non-porous membrane may be transferred (bonded) to one surface of the porous reinforcing material, to form the non-porous membrane on the porous reinforcing material. Among these methods, the former method is preferred from the viewpoint of excellent adhesion between the non-porous membrane and the porous reinforcing material.

[0098] The surface of the porous reinforcing material on the side where the non-porous membrane is to be provided may be subjected to a hydrophilization treatment. Examples of the hydrophilization treatment include those described above.

[0099] The non-porous membrane can be formed by applying (coating) a composition for forming the non-porous membrane on the porous reinforcing material or the other support, and then removing the solvent from the resultant coating membrane by heating or the like.

[0100] Since blocking is suppressed by subjecting the porous reinforcing material to the hydrophilization treatment, the laminate for humidification can be produced by a roll-to-roll process in which a non-porous membrane is formed on the porous reinforcing material produced preliminarily in the form of a wound body, and then the resultant laminate is brought into a wound body again.

[0101] The aforementioned composition can be produced by a known or commonly used method. For example, the composition can be produced by dissolving or dispersing the aforementioned copolymer in a solvent, and mixing the solution or the dispersion with an additive such as a preservative as necessary. The solvent is preferably water and/or a water-soluble solvent. When water or a water-soluble solvent is used, it is presumed that the copolymer is dispersed in the composition in a core-shell form including a hydrophobic moiety on the inner side and a hydrophilic moiety on the outer side. It is presumed that the use of such a composition makes it possible to phase-separate the hydrophilic moiety and the hydrophobic moiety to form a non-porous membrane in a state of having a water-conducting path when the coating film is dried, and the hydrophobic moieties are strongly bonded to each other, whereby the water resistance is further improved.

[0102] Examples of the water-soluble solvent include water-soluble aliphatic alcohols such as methanol, ethanol, n-propanol, and i-propanol; and glycol ethers such as ethylene glycol monomethyl ether, ethylene glycol monobutyl ether, propylene glycol monomethyl ether, and propylene glycol monoethyl ether. One type of the aforementioned water-soluble solvents may be used alone, or two or more types thereof may be used.

[0103] The proportion (concentration) of the copolymer in the composition in which the copolymer is dissolved or dispersed is not particularly limited, but is preferably from 0.5 to 5 mass %, more preferably from 1 to 4 mass %, and still more preferably from 1.5 to 3 mass %. When the concentration is 5 mass % or less, the thickness of the coating layer increases, and thus the thickness of the non-porous membrane after drying becomes more uniform. This makes it possible to form a non-porous membrane having excellent gas barrier properties and a further reduced thickness, resulting in further improved moisture permeability. In addition, when the concentration is within the above range,

it is easy to form a non-porous membrane having excellent coating properties, moisture permeability, and gas barrier properties.

[0104] A known coating method may be used for the application (coating) of the composition. For example, a coater such as a gravure roll coater, a reverse roll coater, a kiss roll coater, a dip roll coater, a bar coater, a knife coater, a spray coater, a comma coater, or a direct coater may be used. The non-porous membrane is readily formed through such a coating method.

[0105] The heating temperature at the time of removing the solvent from the coating membrane is preferably from 35° C. to 90° C., more preferably from 40° C. to 85° C., and still more preferably from 45° C. to 80° C. A proper time can be appropriately adopted for the heating, and the heating time is, for example, from 5 seconds to 20 minutes, preferably from 5 seconds to 10 minutes, and more preferably from 10 seconds to 5 minutes. Since the non-porous membrane can be formed at a low temperature of 90° C. or lower (in particular, 80° C. or lower) using the composition, the membrane is easily formed, and a polyolefin-based resin having excellent moisture permeability can be used as the porous reinforcing material.

[0106] Since the laminate for humidification in which the non-porous membrane is formed on at least one surface of the porous reinforcing material by the above-described method has water repellency and recesses and protrusions, blocking between non-porous membranes can be suppressed, and the non-porous membranes can be disposed close to each other. Thus, the humidifier can be further downsized. The laminate for humidification means a laminate capable of transferring water vapor through the laminate for humidification. Since the laminate for humidification has a structure in which a hydrophilic moiety and a hydrophobic moiety are phase-separated, the laminate for humidification can also be used as a barrier film having moisture permeability. When the diameter of the hydrophilic moiety is small, the barrier film having moisture permeability allows a small hydrophilic substance (for example, water vapor) to pass through the hydrophilic moiety and prevents a large substance (for example, virus) from passing through the hydrophilic moiety, and thus these substances can be separated from each other.

[Bag-Shaped Water-Holding Container]

[0107] A bag-shaped water-holding container can be produced by covering the outside of a frame with the laminate for humidification.

[0108] The frame is preferably made of a rigid material, such as a resin, a metal, or an alloy material. The frame may be a plate-like or bag-shaped frame formed by extrusion molding, a plate-like frame formed by injection molding, or a bag-shaped structure formed by bonding two or more plate-like frames. A hole may be provided after production of the bag-shaped frame, or a frame provided with a hole in advance may be produced.

[0109] It is preferable that a part or the whole of a contact portion between the laminate and the frame is bonded. The method for bonding of the laminate to the frame is not particularly limited, and the bonding can be performed by an adhesion method using a known or commonly used adhesive, or a fusion method such as ultrasonic fusion, high-frequency fusion, or thermal fusion.

[0110] The bag-shaped water-holding container can be produced by the above-described method. In the bag-shaped water-holding container, the laminate is disposed on the outer side of the frame structure. Thus, liquid water passing through the bag-shaped water-holding container is absorbed in the form of water vapor by the laminate and is diffused in the laminate, and the water vapor can be released from the entire laminate.

[0111] A humidifier can be produced by mounting a plurality of the bag-shaped water-holding containers. Since such a humidifier has improved humidification performance as compared with known humidifiers, the humidifier can be reduced in size and can exhibit the comparable performance, and is also excellent in cost.

[0112] Since the humidifier includes the laminate for humidification, the humidifier can be downsized. Thus, the humidifier can be suitably used for applications such as an air conditioner, a ventilator, and an air purifier.

[0113] Each aspect disclosed in the present specification can be combined with any other feature disclosed herein. Each of the configurations, combinations thereof, or the like in each embodiment are an example, and additions, omissions, replacements, and other changes of the configurations may be made as appropriate without departing from the spirit of the present disclosure. In addition, each aspect of the invention according to the present disclosure is not limited by the embodiments or the following examples but is limited only by the claims.

EXAMPLES

[0114] An embodiment of the present disclosure will be described in detail below with reference to Examples.

Example 1

[0115] Silicone (trade name "X-51-1264", available from Shin-Etsu Chemical Co., Ltd.) and silica fine particles (average particle size: 1.7 μm , trade name "SYLSIA 300", available from Fuji Silysia Chemical Ltd.) were added to 100 parts by mass of a thermoplastic resin solution containing 47 parts by mole of isobutyl acrylate and 1 part by mole of styrene as constituent units that form a hydrophobic moiety in a polymer, 21 parts by mole of acrylic acid and 28 parts by mole of methyl acrylate as constituent units that form a hydrophilic moiety, 3 parts by mole of an imide as a constituent unit that forms a cationic moiety, and 2 parts by mole of an epoxy compound as a crosslinking agent, and the resultant mixture was diluted with 21 parts by mass of ethanol and 154 parts by mass of distilled water, to prepare a coating liquid. Meanwhile, one surface of a polypropylene-made porous base material (thickness: 20 μm , porosity: 48 vol %) was subjected to a corona treatment to form a hydrophilic surface having a surface tension of 39 dyn. Then, the coating liquid was applied to the hydrophilic surface of the porous base material using a wire bar and heated at 70° C. for 1 minute to form a non-porous membrane (thickness: 1 μm). Thus, a laminate for humidification of Example 1 was produced. In the laminate for humidification of Example 1, the average particle size of the silica fine particles was larger than the thickness of the non-porous membrane, and the non-porous membrane was provided with recesses and protrusions by the silica fine particles and provided with water repellency by silicone particles.

Examples 2 and 3 and Comparative Examples 1 and 2

[0116] A laminate for humidification was produced in the same manner as in Example 1, except that the components marked with o were added as shown in Table 1 to prepare a coating liquid. Each of the produced laminates for humidification has recesses and protrusions as shown in Table 1.

[0117] Details of the respective components in Table 1 other than those described above are described below.

[0118] Silica fine particle B: trade name “MP-4540M”, available from Nissan Chemical Corporation, average particle size: 0.45 μm

[0119] Chitosan: trade name “EFo-08002”, available from Sugino Machine Ltd.

[0120] Since chitosan is in the form of nanofibers rather than particles, the average particle size cannot be described. No problem occurs if this is deleted from the Examples, so please correct the description in a preferred way.

[Evaluation]

[0121] Each of the laminates for humidification produced in Examples and Comparative Examples was evaluated as follows. The evaluation results are shown in the tables.

(1) Water Contact Angle

[0122] Using a contact angle measuring device (trade name “DropMaster DMs-401” available from Kyowa Interface Science Co., Ltd.), about 2 μL of distilled water was added dropwise onto a surface of the non-porous membrane of the produced laminate for humidification, and the angle (contact angle) between the droplet and the surface was measured at three points. The average of the measured values was used as a water contact angle for evaluation.

(2) Arithmetic Average Roughness (Sa)

[0123] In accordance with JIS B 0681, the arithmetic average height (Sa) was determined based on a range obtained through measurement of recesses and protrusions on the surface of the non-porous membrane by using a non-contact surface/layer cross-sectional shape measurement system (trade name: “VertScan 2.0”, available from Ryoka Systems Inc.) and an objective lens with a magnification of 5 $\times$ in a field of view of 2.51 mm $\times$ 1.88 mm.

(3) Water Retention Amount

[0124] The laminate for humidification was cut into a circular test piece having a diameter of 7 cm, and the non-porous membrane side of the test piece of the laminate was impregnated with pure water for 15 minutes and then taken out. An increment from the mass of the test piece of the laminate for humidification before the test was converted into a mass per 1 m² of test piece and calculated as a water retention amount.

(4) Anti-Blocking Property

(Evaluation in Roll Form)

[0125] The laminate for humidification in a roll form was heated at 70° C. for 15 hours, and then the laminate for humidification was pulled out by hand. The blocking property was evaluated as follows. In this test, anti-blocking between the non-porous membrane and the porous base

material in the laminate for humidification was evaluated, and the laminate exhibiting Medium or Low was evaluated as having an anti-blocking property.

[0126] High: The laminate for humidification cannot be pulled out.

[0127] Medium: The laminate for humidification is deformed after the laminate for humidification is pulled out.

[0128] Low: The laminate for humidification is not deformed at all after the laminate for humidification is pulled out.

(Evaluation in Sheet Form)

[0129] The laminate for humidification was prepared into two strip samples each having a size of 25 mm $\times$ 100 mm. Water was added dropwise to the non-porous membrane of one sample, and then the non-porous membrane was bonded to the non-porous membrane of the other laminate for humidification, whereby the entire bonded surface was impregnated with water. The bonded sample was heated at 55° C. for 1 hour and the peel strength was measured with Tensilon. In this test, anti-blocking between the surfaces of non-porous membranes in the laminate for humidification was evaluated, and a sample exhibiting a peel strength of 3 N/25 mm or less was evaluated as having an anti-blocking property.

TABLE 1

	Example			Comparative Example	
	1	2	3	1	2
Silica fine particle A	○				
Silica fine particle B		○			
Silicone	○	○			○
Chitosan			○		
Water contact angle	97	95	92	74	93
Arithmetic average roughness (Sa)	0.23	0.18	0.43	0.10	0.08
Water retention amount	8.7	—	36.1	38.0	38.0
Anti-blocking property in roll form	Low	Medium	Low	High	High
Anti-blocking property in sheet form	0.1	1.5	0.1	3.3	3.8

[0130] As shown in Examples 1 to 3, it was confirmed that blocking between non-porous membranes was suppressed in the laminate for humidification in which the surface of the non-porous membrane had water repellency and recesses and protrusions. Blocking was confirmed in the laminate for humidification (Comparative Example 1) having insufficient water repellency and small recesses and protrusions, or in the laminate for humidification (Comparative Example 2) having only water repellency and small recesses and protrusions.

[0131] Hereinafter, variations of the invention according to the present disclosure will be described.

[Appendix 1]

[0132] A laminate for humidification, the laminate including a porous reinforcing material, and a non-porous membrane that is laminated on at least one surface of the porous

reinforcing material and functions as a moisture-permeable membrane, wherein a surface of the non-porous membrane has water repellency and recesses and protrusions.

[Appendix 2]

[0133] The laminate for humidification according to Appendix 1, wherein the non-porous membrane contains an anti-blocking agent.

[Appendix 3]

[0134] The laminate for humidification according to Appendix 2, wherein the anti-blocking agent has an average particle size larger than a thickness of the non-porous membrane, and the recesses and protrusions are formed by the anti-blocking agent.

[Appendix 4]

[0135] The laminate for humidification according to any one of Appendices 1 to 3, wherein the surface of the non-porous membrane has a water contact angle of from 90° to 120°.

[Appendix 5]

[0136] The laminate for humidification according to any one of Appendices 1 to 4, wherein the surface of the non-porous membrane has an arithmetic average roughness (Sa) of from 0.1 μm to 0.5 μm .

[Appendix 6]

[0137] The laminate for humidification according to any one of Appendices 1 to 5, wherein the non-porous membrane is disposed on a liquid water side of the porous reinforcing material and is used to absorb liquid water and release the liquid water into air.

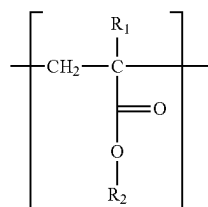
[Appendix 7]

[0138] The laminate for humidification according to any one of Appendices 1 to 6, wherein the non-porous membrane contains a thermoplastic resin including a cationic moiety, and the cationic moiety contains a group containing an ammonium ion or a group capable of forming an ammonium ion.

[Appendix 8]

[0139] The laminate for humidification according to any one of Appendices 1 to 7, wherein the non-porous membrane contains a thermoplastic resin including a hydrophilic moiety, and the hydrophilic moiety contains a constituent unit represented by Formula (1) described below:

[Chem. 1]

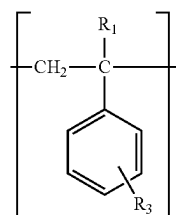


[0140] (wherein R_1 and R_2 each independently represent a hydrogen atom or a methyl group).

[Appendix 9]

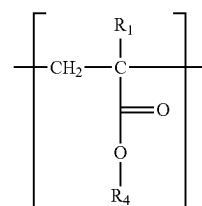
[0141] The laminate for humidification according to any one of Appendices 1 to 8, wherein the non-porous membrane contains a thermoplastic resin including a hydrophobic moiety, and the hydrophobic moiety contains a constituent unit represented by Formula (2) and/or Formula (3) described below:

[Chem. 2]



[0142] (wherein R_1 represents a hydrogen atom or a methyl group, and R_3 represents an alkyl group having from 1 to 2 carbons)

[Chem. 3]



[0143] (wherein R_1 represents a hydrogen atom or a methyl group, and R_4 represents a branched alkyl group having 3 or more carbons).

[Appendix 10]

[0144] The laminate for humidification according to any one of Appendices 1 to 9, wherein the laminate for humidification has a water retention amount of from 5 g/m^2 to 30 g/m^2 .

[Appendix 11]

[0145] The laminate for humidification according to any one of Appendices 1 to 10, wherein the non-porous membrane is coated to cover at least one surface of the porous reinforcing material.

[Appendix 12]

[0146] A humidifier including a plurality of bag-shaped water-holding containers each including the laminate for humidification described in any one of Appendices 1 to 11 and a frame, the laminate and the frame being bonded together.

[Appendix 13]

[0147] An air conditioner including the humidifier described in Appendix 12.

[Appendix 14]

[0148] A ventilator including the humidifier described in Appendix 12.

[Appendix 15]

[0149] An air purifier including the humidifier described in Appendix 12.

REFERENCE SIGNS LIST

- [0150] 1 Laminate
- [0151] 11 Porous reinforcing material
- [0152] 11a One surface of porous reinforcing material
- [0153] 12 Non-porous membrane

1. A laminate for humidification, the laminate comprising a porous reinforcing material, and a non-porous membrane that is laminated on at least one surface of the porous reinforcing material and functions as a moisture-permeable membrane, wherein a surface of the non-porous membrane has water repellency and recesses and protrusions.

2. The laminate for humidification according to claim 1, wherein the non-porous membrane contains an anti-blocking agent.

3. The laminate for humidification according to claim 2, wherein

the anti-blocking agent has an average particle size larger than a thickness of the non-porous membrane, and

the recesses and protrusions are formed by the anti-blocking agent.

4. The laminate for humidification according to claim 1, wherein the surface of the non-porous membrane has a water contact angle of from 90° to 120°.

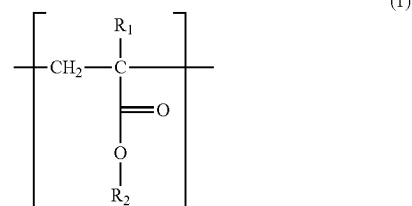
5. The laminate for humidification according to claim 1, wherein the surface of the non-porous membrane has an arithmetic average roughness (Sa) of from 0.1 μm to 0.5 μm.

6. The laminate for humidification according to claim 1, wherein the non-porous membrane is disposed on a liquid water side of the porous reinforcing material and is used to absorb liquid water and release the liquid water into air.

7. The laminate for humidification according to claim 1, wherein the non-porous membrane contains a thermoplastic resin including a cationic moiety, and the cationic moiety contains a group containing an ammonium ion or a group capable of forming an ammonium ion.

8. The laminate for humidification according to claim 1, wherein the non-porous membrane contains a thermoplastic resin including a hydrophilic moiety, and the hydrophilic moiety contains a constituent unit represented by Formula (1):

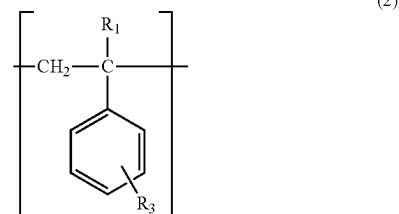
[Chem. 1]



wherein R₁ and R₂ each independently represent a hydrogen atom or a methyl group.

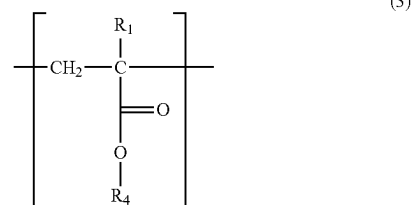
9. The laminate for humidification according to claim 1, wherein the non-porous membrane contains a thermoplastic resin including a hydrophobic moiety, and the hydrophobic moiety contains a constituent unit represented by Formula (2) and/or Formula (3):

[Chem. 2]



wherein R₁ represents a hydrogen atom or a methyl group, and R₃ represents an alkyl group having from 1 to 2 carbons,

[Chem. 3]



wherein R₁ represents a hydrogen atom or a methyl group, and R₄ represents a branched alkyl group having 3 or more carbons.

10. The laminate for humidification according to claim 1, wherein the laminate for humidification has a water retention amount of from 5 g/m² to 30 g/m².

11. The laminate for humidification according to claim 1, wherein the non-porous membrane is coated to cover at least one surface of the porous reinforcing material.

12. A humidifier comprising a plurality of bag-shaped water-holding containers each including the laminate for humidification described in claim 1 and a frame, the laminate and the frame being bonded together.

13. An air conditioner comprising the humidifier described in claim 12.

14. A ventilator comprising the humidifier described in claim **12**.

15. An air purifier comprising the humidifier described in claim **12**.

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